USER INFORMATION

INJECSOL S20

SODIUM CHLORIDE 20% w/v INJECTION BP

Sterile and Non Pyrogenic Concentrate

Composition:
Each 10 mL concentrate contains:

Sodium Chloride	2 g
Sodium Ion (Na^+)	34.2 mmol
Chloride Ion (Cl^-)	34.2 mmol

Product Description:
INJECSOL S20 is a clear, colourless, sterile and non-pyrogenic concentrated solution for infusion.

Pharmacodynamics:
The principal determinant of the effective osmolality of the extracellular fluids (and also of the intracellular fluids, since they remain in osmotic equilibrium with the extracellular fluids) is the extracellular fluid Sodium concentration. The reason for this is that sodium is the most abundant positive ion of the extracellular fluid. Negative ion concentrations of the body fluids are adjusted to equal those of the positive ions by renal acid-base control mechanisms. Furthermore, Glucose and urea, the most abundant of the non-ionic osmolar solutes in extracellular fluids, normally only represent about 3% of the total osmolality. Therefore, in effect, the extracellular fluid Sodium ion concentration controls over 90% of the effective osmotic pressure of the extracellular fluid. Sodium Chloride remains the most important single salt for prophylaxis or replacement therapy of deficits of extracellular fluid. Volume contraction, whether isotonic, hypotonic or hypertonic, may seriously impair the circulation (cardiac output falls and microcirculation is compromised).

Pharmacokinetics:
As the Sodium Chloride Intravenous preparations are directly administered to the circulation, the bioavailability of the components is 100%. Excess sodium is predominantly excreted by the kidneys, with small amounts lost in faeces and sweat.

Indication:
- Electrolyte rebalance when small supplementation of water is desired, in slow intravenous infusion.
- Sodium supplementation with reduced volume in solutions for parenteral nutrition.

Recommended Dosage:
The dose is adjusted according to the analytical values of the serum ionogram and should also be adjusted according to the analytical values of the acid-base status.

The Sodium deficit is calculated by the formula:
Sodium deficit (mmol) = ($Na^+_{required}$ – Na^+_{actual}) x kg b.w. x 0.2
(The extracellular volume is calculated as body weight x 0.2)

Maximum Daily Dose
The maximum daily dose depends on the actual need of electrolytes. As a rule, the daily dose for adults is 3 - 6 mmol/kg b.w., for children 3 - 5 mmol/kg b.w.
Maximum Infusion Rate
The maximum infusion rate depends on the prevailing clinical situation.

Method of Administration
Intravenous use, only diluted by addition to a suitable infusion solution.
In general, the calculated amount of sodium chloride is added to 250 mL of fluid. In cases of fluid deficit larger volumes of vehicle solution may be used. For infusion into peripheral veins, the solution must be diluted so as not to exceed an osmolarity of 800 mOsm/L.
Care should be taken to add the sodium chloride concentrate to the infusion solution under strictly aseptic conditions immediately before setting up the infusion. After mixing the infusion bottle should be gently shaken.

Route of Administration: Intravenous(IV)

Contraindications:
INJECSOL S20 must not be used in cases of:
- Hypernatraemia
- Hyperchloraemia

INJECSOL S20 should only be administered with caution in cases of:
- Hypokalaemia,
- Disorders where restriction of sodium intake is indicated, such as cardiac insufficiency, generalised oedema, pulmonary oedema, hypertension, eclampsia, severe renal insufficiency,
- Therapy with corticosteroids or with ACTH,
- Metabolic acidosis.

Warnings and Precautions:
Warnings:
INJECSOL S20 is a concentrated solution and **MUST BE DILUTED BEFORE USE** by adding to a large volume intravenous fluid before use.

INJECSOL S20 MUST NOT BE INJECTED UNDILUTED.

Sodium Chloride Concentrate should be administered with considerable care to patients with cardiac insufficiency, peripheral or pulmonary oedemas or severe renal impairment. The intravenous administration of this solution (after appropriate dilution) can cause fluid and/or solute overloading resulting in dilution of other serum electrolyte concentrations, overhydration, congested states or pulmonary edema.
Excessive administration of potassium free solutions may result in significant hypokalemia.
Inadvertent direct injection or absorption of concentrated sodium chloride solution may give rise to sudden hypernatremia and such complications as cardiovascular shock, central nervous system disorders, extensive haemolysis, cortical necrosis of the kidneys and severe local tissue necrosis (if administered extravascularly).
Solutions containing sodium ions should be used with great care, if at all, in patients with congestive heart failure, severe renal insufficiency and in clinical states in which there exists edema with sodium retention.
In patients with diminished renal function, administration of solutions containing sodium may result in sodium retention.
This product contains aluminum that may be toxic. Aluminum may reach toxic levels with prolonged parenteral administration if kidney function is impaired. Premature neonates are particularly at risk because their kidneys are immature, and they require large amounts of calcium and phosphate solutions which contain aluminum. Research indicates that patients with impaired kidney function, including premature neonates, who receive parenteral levels of aluminum at greater than 4 to 5 mcg per kg per day accumulate aluminum at levels associated with central nervous system and bone toxicity. Tissue loading may occur at even lower rates of administration of TPN products and of the lock-flush solutions used in their administration.
The administration to neonates and elderly patients should be closely monitored. The therapy must be performed under strict medical monitoring, with dose adjustment depending on the result of electrolyte assessments. The solution being hypertonic, sodium rebalance should be performed slowly.
Sodium Chloride solutions should not be used to induce emesis; this practice is dangerous and deaths from resulting hypernatraemia have been reported.
Precautions:
Clinical monitoring should include checks of the serum electrolytes, the water balance, and the acid-base balance.
Special caution should be used in administering sodium-containing solutions to patients with severe renal impairment, cirrhosis of the liver, cardiac failure, or other edematous or sodium- retaining states.

Caution must be exercised in the administration of parenteral fluids, especially those containing Sodium ions, to patients receiving corticosteroids or corticotropin.
INJECSOL S20 must be used immediately after dilution. The compatibility of the large volume IV fluid intended for dilution should be checked before use. Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Interactions with Other Medicaments:
During therapy with corticosteroids or ACTH there may be an increased retention of Sodium and Chloride. Incompatibility can arise upon mixing with other medicaments. The attending doctor will decide on the use of mixed infusions.

Pregnancy and Lactation:
Use in Pregnancy
Safety in pregnancy has not been established. Use is recommended only when clearly indicated.
Use in Lactation
Safety in lactation has not yet been established. Use of this product while breastfeeding is recommended only when potential benefits outweigh potential risks to the newborn infant.

Side Effects:
Adverse effects of Sodium salts are attributable to electrolyte imbalances from excess Sodium. Retention of excess Sodium in the body usually occurs when there is defective renal Sodium excretion. This leads to the accumulation of extracellular fluid to maintain normal plasma osmolality, which may result in pulmonary and peripheral oedema and their consequent effects. Hypernatraemia (a rise in plasma osmolality) is usually associated with inadequate water intake, or excessive water losses . The most serious effect of hypernatraemia is dehydration of the brain which causes somnolence and confusion progressing to convulsions, coma, respiratory failure, and death. Other symptoms include thirst, reduced salivation and lachrymation, fever, sweating, tachycardia, hypertension or hypotension, headache, dizziness, restlessness, irritability, weakness, and muscular twitching and rigidity. Administration of larger amounts of the solution may lead to hypernatraemia and hyperchloraemia. Patients are advised to inform their doctor or pharmacist of any adverse reaction they experience in connection with the administration of this drug.

Symptoms and Treatment of Overdose:
Symptoms:
Excess Sodium Chloride in the body produces general gastrointestinal effects of nausea, vomiting, diarrhoea and cramps. Salivation and lacrimation are reduced, while thirst and sweating are increased. Hypotension, tachycardia, renal failure, peripheral and pulmonary oedema and respiratory arrest may occur. CNS symptoms include headache, dizziness, restlessness, irritability, weakness, muscular twitching and rigidity, convulsions, coma and death.

Treatment:
Normal plasma Sodium concentrations should be carefully restored at a rate not greater than 10-15 mmol/day using intravenous hypotonic saline.
In mild cases, oral administration of water and restriction of Sodium intake is sufficient. In more severe cases, dialysis may be necessary if there is significant renal impairment, the patient is moribund or plasma Sodium levels are greater than 200 mmol/L. Convulsions are to be treated with intravenous Diazepam.

Storage Condition: Do not store above 30°C.

Shelf life:
2 years from manufacturing date in proposed storage condition in LDPE ampoule. Do not use after expiry.

Dosage form and packaging available:
10 mL X 20 LDPE plastic ampoules per box

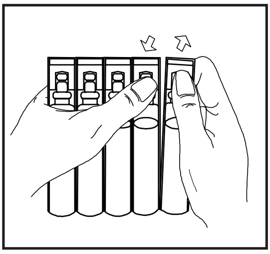
Manufacturer/ Product Registration Holder:

AIN MEDICARE SDN. BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44,
Kaw. Perindustrian Pengkalan Chepa 2,
16100 Kota Bharu, Kelantan, MALAYSIA

Date of Revision : 01.04.2024

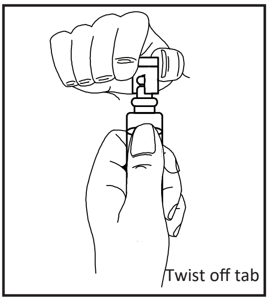
HANDLING INSTRUCTION
AIN MEDICARE INJECTION FLUID IN POLYETHYLENE (PE) AMPOULE

1)



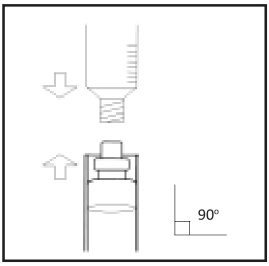
Detach ampoule along parting line

2)



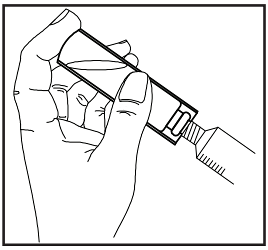
Twist off tab

3)



Attach luer lock syringe, luer slip syringe or hypodermic needle vertically to the ampoule.

4)



Draw out injection fluid using luer lock syringe, luer slip syringe or hypodermic needle. Ensure the connection is tightly secured for the liquid.